SemanticSplat: Graph-Pruned Semantic Search for Queryable 3DGS Digital Twins

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Abstract. Built-environment digital twins based on 3D Gaussian Splatting (3DGS) are photorealistic but lack an explicit semantic layer for natural-language querying. We present SemanticSplat, a graph-pruned semantic search layer that organizes captured views into a hierarchical zone–region–object tree and answers queries via top-down traversal instead of exhaustive flat search over the same semantic items. On five manually captured indoor/outdoor digital-twin pilot scenes (96 views; 150 queries), graph traversal cuts average views checked from 19.4 to 4.77 (75.5%) and input tokens from 3168 to 1014 (68.2%). Under stub lexical matching with verified view labels (n=125), hit@1 is 0.68 (graph) vs. 0.768 (flat), so the internal track shows a quality–cost tradeoff rather than free quality preservation. We further align evaluation with BBQ-style object grounding on official Replica and ScanNet pilot subsets: on a 48-query ScanNet language pilot, graph search matches flat lexical Recall@1 / Acc@0.25 (0.271) while checking about 77% fewer objects. We position the work as hierarchical context pruning for queryable 3DGS twins, not as a claim of superiority over BBQ, whose official code is not reproduced here.

Keywords: Digital twins · 3D Gaussian Splatting · Scene graphs · Semantic search · 3D object grounding.

1 Introduction

Digital twins of the built environment increasingly rely on neural rendering, including 3D Gaussian Splatting [3], for interactive visualization. TwinWorld-relevant applications—facility queries, robotics hand-off, and semantic mapping—need more than geometry and appearance: a user should ask "where can I sit near the reception?" and receive a grounded viewpoint or object. Exhaustive flat search over all views or objects is simple but wastes context and scales poorly with scene size.

Language-field methods [4, 6] distill features into the field but do not expose hierarchical pruning. Object-centric scene graphs such as ConceptGraphs [1] and BBQ [5] provide strong public-dataset object grounding; BBQ in particular reports Acc@k on Replica/ScanNet and Sr3D+/Nr3D/ScanRefer. Our focus is complementary: a hierarchical semantic index over 3DGS-captured views with measured *query-time* context pruning, evaluated under a same-input graph-vs-flat protocol and aligned with BBQ-style Acc@k / Recall@1 on public-dataset pilots.

Contributions.

- A rendering-first semantic-map layer for queryable 3DGS digital twins with hierarchical traversal and affordance expansion.
- Same-input graph-vs-flat evidence on five captured scenes and BBQ-aligned object-grounding pilots on official Replica (8 scenes) and ScanNet (8 scenes).
- Honest baseline status: internal flat lexical / embedding variants, ConceptGraphs full captured-scene run, LangSplat smoke, BBQ as related-work metric guide (official BBQ code not run).

2 Related Work

Language fields and 3DGS semantics. LERF [4], LangSplat [6], and related Gaussian semantics embed language for open-vocabulary pointing. They optimize a different native representation than tree traversal over shared semantic items.

Object-centric and hierarchical scene graphs. ConceptGraphs [1], BBQ [5], HOV-SG [7], and Hydra [2] construct graphs for grounding, navigation, or robotics. BBQ is the closest public-dataset grounding reference: object IDs, captions, 3D extents, spatial relations, and deductive LLM reasoning over a compact subgraph [5]. We cite BBQ Acc@k tables as literature context, not as measurements from this repository.

Positioning. SemanticSplat targets hierarchical context control for 3DGS digital twins. Measured comparisons are same-input graph vs. flat lexical/embedding on identical scenes, items, queries, and labels. We do not claim superiority over BBQ without a same-protocol official-code run.

3 Method

3.1 Capture and semantic index

Users capture posed RGB(-D) views from a 3DGS or RGB-D scene and annotate or convert structured view records (objects, landmarks, room cues, short summaries). These records form the shared search units for both graph and flat methods (Fig. 1).

3.2 Hierarchical tree and traversal

Items are organized as zones $\rightarrow$ regions $\rightarrow$ object/view leaves. Queries are decomposed into constraints; the tree is traversed top-down and irrelevant branches are pruned before leaf confirmation (Fig. 2). The flat baseline scans the same items exhaustively.

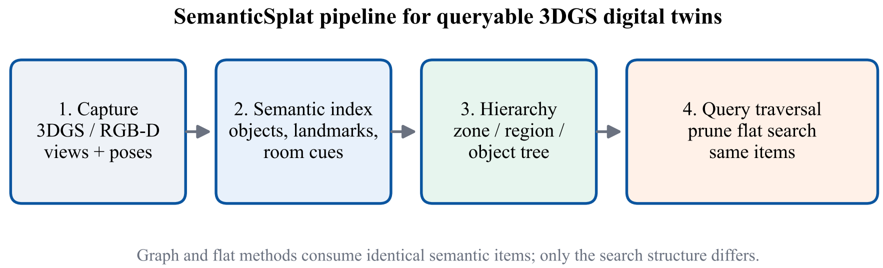

Fig. 1: SemanticSplat pipeline: capture → semantic index → hierarchy → query-time pruning. Graph and flat consume the same items.

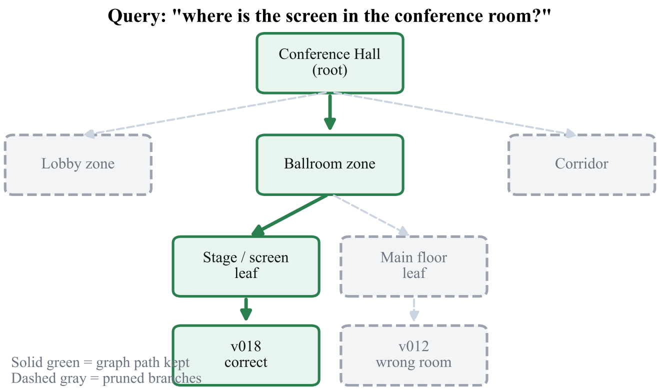

Fig. 2: Schematic top-down traversal for a room-scoped screen query: solid path kept, dashed branches pruned.

3.3 Object grounding output

For BBQ-aligned evaluation, each query records selected object/view/node IDs, predicted 3D bbox when available, traversal path, checked objects/views/nodes, runtime, and estimated tokens. Construction cost is reported separately from query-time cost.

3.4 Search variants

We evaluate graph, graph+calibrated fallback, flat lexical, and flat embedding on the same public-dataset pilots, plus ablations (no hierarchy, no relation rerank, no fallback).

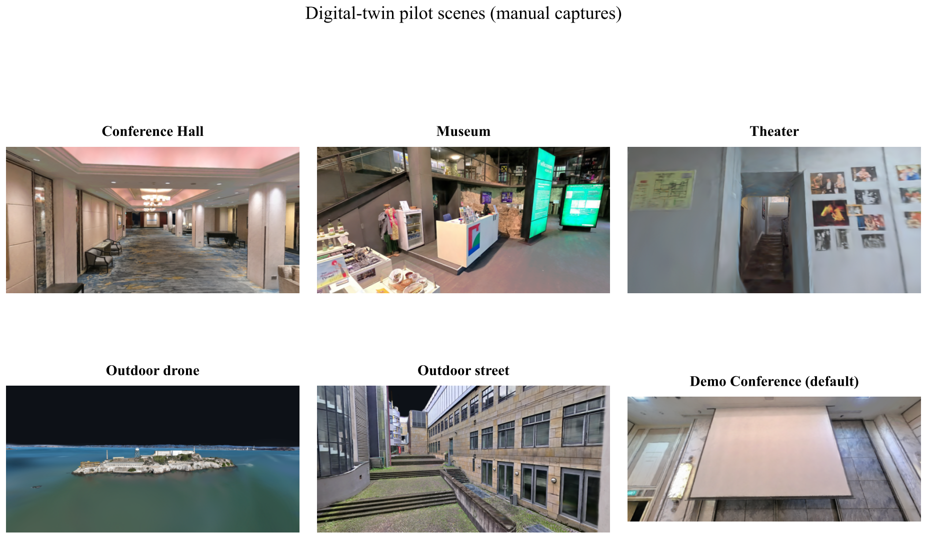

Fig. 3: Captured digital-twin pilot scenes used in the internal five-scene track.

4 Datasets and Evaluation

4.1 Internal digital-twin pilots

Five manually captured scenes (Conference Hall, Museum, Theater, outdoor drone/street; 96 views) provide digital-twin prototype evidence (Fig. 3). Metrics: views checked, input tokens, hit@k on verified view labels. Manual annotations are *reference labels*, not independent public GT.

4.2 Public-dataset pilots (Person 1+2)

We use the BBQ scene subset where possible: Replica room0–2 and office0–4; ScanNet 0011_00, 0030_00, 0046_00, 0086_00, 0222_00, 0378_00, 0389_00, 0435_00. Pilot tracks:

- Replica: 56 queries, 8 scenes, official GT object boxes (oracle semantic candidates).
- ScanNet: 48 official Nr3D/Sr3D+ queries, 8 scenes.

BBQ-aligned metrics: Recall@1, Acc@0.1 / 0.25 / 0.5. Segmentation mAcc/mIoU/fmIoU are **N/A** (no paired predicted segmentation arrays in this track).

5 Experiments

5.1 Internal five-scene graph vs. flat

Across 150 queries, graph traversal reduces average views checked from 19.4 to 4.77 (75.5%) and input tokens from 3168 to 1014 (68.2%). Cumulative tokens are 475k flat vs. 152k graph (Fig. 5). On 125 GT queries with verified

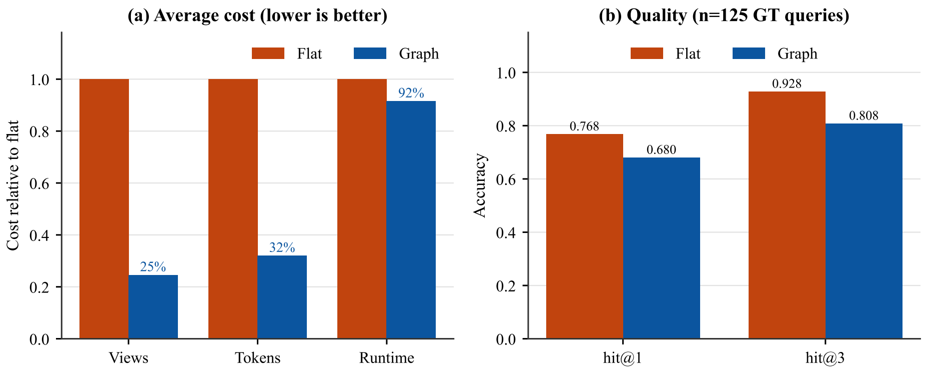

Fig. 4: Internal five-scene cost (normalized to flat) and quality (stub lexical; frozen `metrics_summary.json`).

`expected_view_ids`, stub-lexical hit@1 is 0.68 (graph) vs. 0.768 (flat) and hit@3 is 0.808 vs. 0.928 (Fig. 4). Thus hierarchy buys large context savings with a hit@k tradeoff under the current stub matcher—not a free win on retrieval quality.

5.2 Where hierarchy helps

Per-query-type and per-scene analysis (Figs. 6–7) show larger savings when queries localize to one zone (e.g. negatives, simple objects) and smaller savings for multi-hop and flatter outdoor trees.

5.3 Public-dataset BBQ-aligned pilots

Table 1 summarizes frozen Person 2 runs. On Replica, graph and flat lexical reach a Recall@1 / Acc@k ceiling of 1.0 on oracle candidates—a protocol sanity check, *not* independent perception accuracy. Graph checks 12.1 objects vs. 71.9 for flat lexical. On the harder ScanNet language pilot, graph matches flat lexical quality (Recall@1 = Acc@0.25 = 0.271) while checking 11.2 vs. 49.0 objects ($\approx 77\%$ fewer). Relation reranking does not help this pilot (no-relation Acc@0.25 = 0.292). Figure 8 visualizes the quality/cost trade-off; Figure 9 shows Replica oracle search cost under a Acc@k=1.0 ceiling.

5.4 Case studies

Room-scoped case studies on the Conference Hall demo (Figs. 10–11) illustrate matched correctness at far lower cost for isolated targets, near-flat cost when the answer is zone-wide, and cheap shared misses when lexical retrieval fails for both methods. Figure 11 shows a screen query with the preferred stage view (v018) versus a distractor floor view (v012).

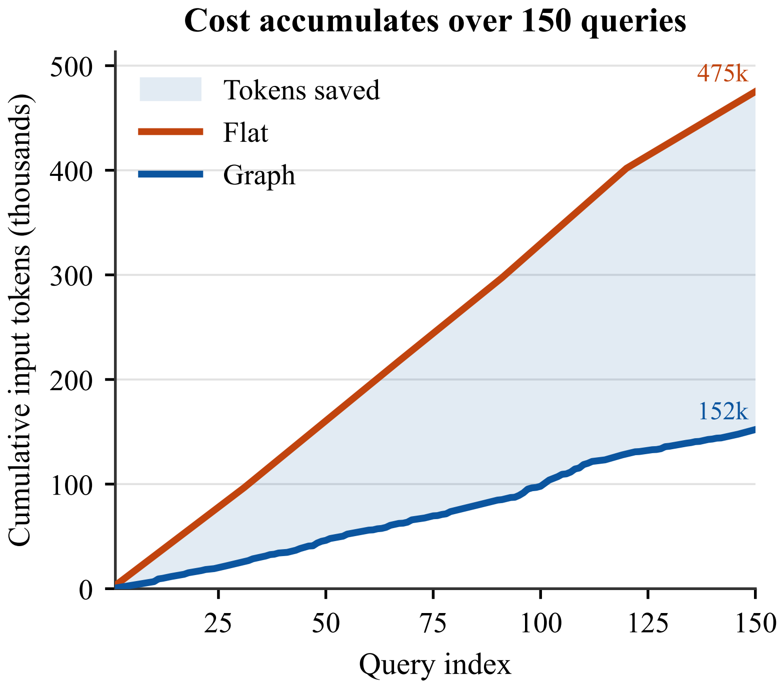

Fig. 5: Cumulative tokens over 150 internal queries (475k flat vs. 152k graph).

5.5 Baselines status

ConceptGraphs has a full five captured-scene run (not Replica/ScanNet). LangSplat has official sofa smoke only. BBQ official code: **not run**; BBQ paper tables are related-work only [5].

6 Limitations

- Manual five-scene annotations are reference labels, not independent public GT.
- Internal stub-lexical hit@k currently favors flat over graph (hit@1 0.68 vs. 0.768); public pilots match quality while cutting objects checked.
- Public pilots use oracle semantic candidates; Replica Acc@k=1.0 is a ceiling sanity result.
- mAcc/mIoU/fmIoU are unavailable without predicted segmentation arrays.
- Relation reranking currently underperforms its ablation on ScanNet pilot.
- No same-protocol BBQ official-code comparison; no superiority claim.
- Stub / deterministic lexical matching is not live provider-backed VLM evaluation.

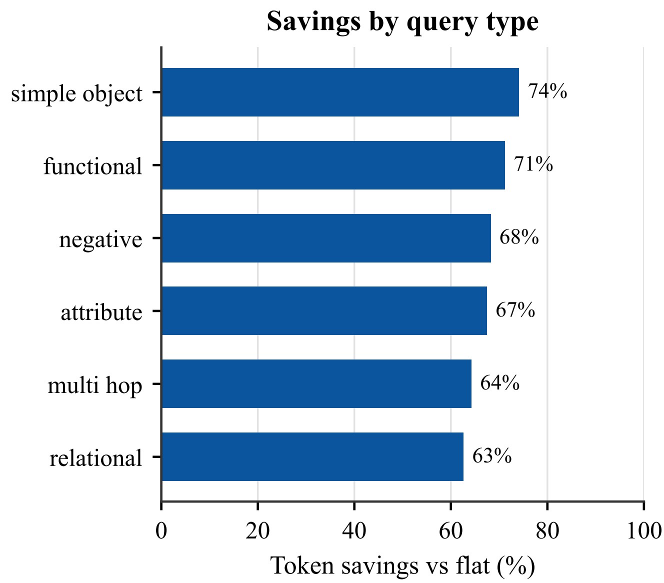

Fig. 6: Token savings by query type on the internal five-scene track.

7 Reproducibility

Frozen paths and the git commit hash are listed in `docs/reports/final/twinworld_rep` Internal five-scene evidence sits under `outputs/graph_vs_flat/five_scene_graph_vs_` public pilots under `outputs/public_datasets/`. Regenerate figures with `export_paper_` / `export_twinworld_figures.py`; gate numbers with `check_twinworld_numbers.py`. Numbers in this PDF follow the frozen JSON (not superseded older notes).

8 Conclusion

SemanticSplat shows that hierarchical graph pruning reduces query-time context for 3DGS digital twins under a controlled same-input protocol. On the internal five-scene track the savings are large (75.5% views / 68.2% tokens) with a stub-lexical hit@k tradeoff; on BBQ-aligned ScanNet/Replica pilots, Acc@k matches flat lexical while checked objects drop sharply. We do not claim BBQ superiority. Future work: stronger relation-aware search, predicted segmentation track, and fair same-protocol BBQ execution if permitted.

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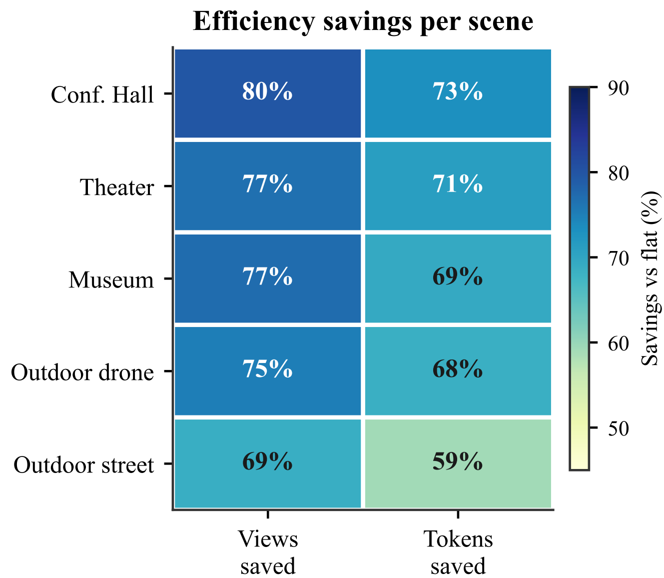

Fig. 7: Per-scene view/token savings heatmap (internal track).

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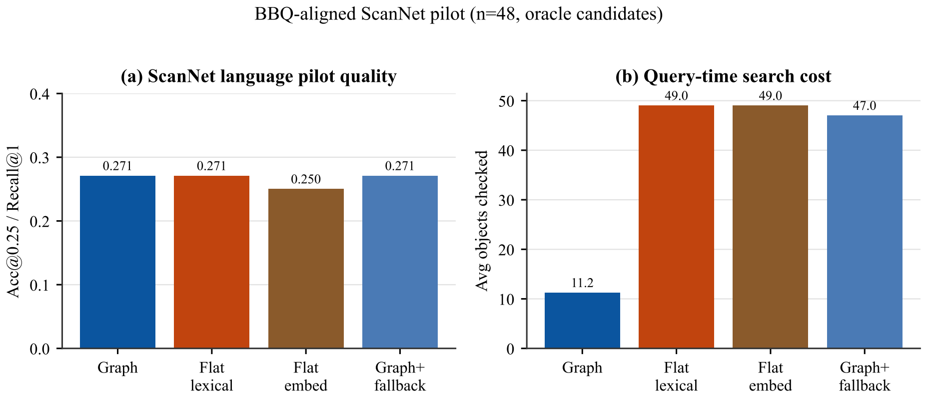

Fig. 8: ScanNet BBQ-aligned pilot (n=48 Nr3D/Sr3D+ queries, oracle candidates). Graph preserves flat-lexical Acc@0.25 while checking far fewer objects; fallback approaches flat cost.

Table 1: Public-dataset pilots (oracle semantic candidates). Efficiency = avg checked objects.

Track	Variant	Recall@1	Acc@0.25	Objects	Tokens
Replica (n=56)	graph	1.00	1.00	12.1	303
	flat lexical	1.00	1.00	71.9	1765
	flat embedding	0.83	0.88	71.9	1765
ScanNet (n=48)	graph	0.271	0.271	11.2	501
	graph+fallback	0.271	0.271	47.0	2027
	flat lexical	0.271	0.271	49.0	2117
	flat embedding	0.250	0.250	49.0	2117

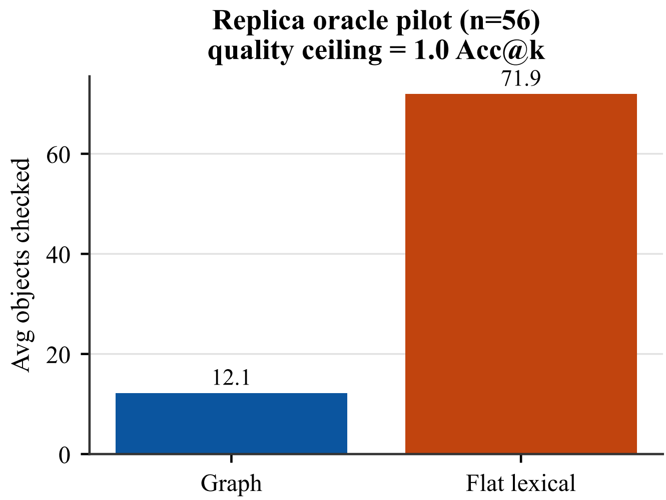

Fig. 9: Replica oracle pilot (n=56): quality ceiling Acc@k=1.0 for graph and flat lexical; graph still reduces objects checked. Not independent perception accuracy.

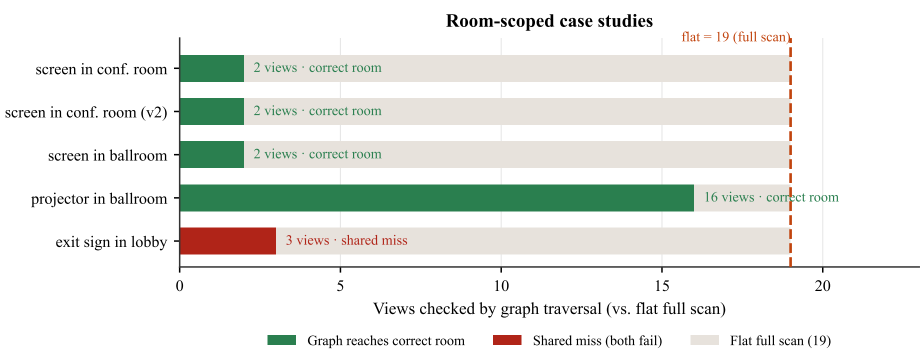

Fig. 10: Room-scoped case studies: graph views checked vs. flat full scan.

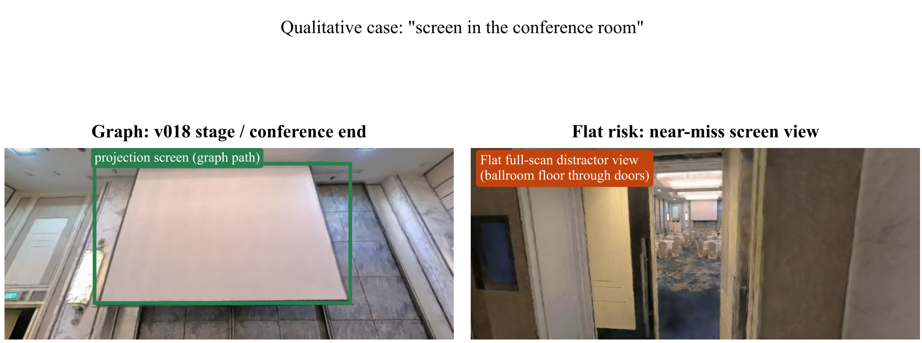

Fig. 11: Qualitative demo view pair for a conference-screen query (bbox from view annotation).